

Unit 1 : Introduction to Computer Networks

❖ **Introduction to data communication**

Data communication refers to the exchange of data between two or more devices through a transmission medium such as wires, optical fiber, or wireless signals.

It involves the transmission of information from a sender to a receiver in a reliable and efficient manner using agreed-upon rules called protocols.

Example: Sending emails, video calls, online banking, file transfer, etc.

○ **Computer Networks**

A computer network is a collection of interconnected devices such as computers, servers, printers, routers, and smartphones that communicate with each other to share data, information, and resources.

Networks allow users to exchange information quickly and efficiently, share hardware and software, and access remote services.

- Objectives of Computer Networks
 - Resource sharing (printers, storage, internet)
 - Fast communication (email, video conferencing)
 - High reliability and backup
 - Cost reduction
 - Centralized management

- Applications

Online banking, e-commerce, cloud services, online classes, file sharing, social networking.

○ **Characteristics**

For data communication to be effective and reliable, it must possess the following characteristics:

1. Delivery: Data must be delivered to the correct destination and user.
2. Accuracy: Data must be delivered without errors. Any alteration or loss of data reduces reliability.
3. Timeliness: Data must be delivered at the correct time. Late data is useless in real-time applications like video conferencing.
4. Jitter: Jitter refers to the variation in packet arrival time. Low jitter is required for smooth audio and video communication.

○ **Components**

A data communication system consists of five basic components:

1. Sender: The device that generates and sends the data (computer, smartphone, server).
2. Receiver: The device that receives the data.
3. Message: The information to be communicated (text, image, audio, video, numbers).
4. Transmission Medium: The physical or wireless path through which data travels. Examples: twisted pair cable, optical fiber, radio waves.
5. Protocol: A set of rules that governs data communication such as formatting, timing, error control, and security. Examples: TCP/IP, HTTP, FTP, SMTP.

○ **Data Representation**

Data representation is the method used by computers to encode information such as text, numbers, images, audio, and video into binary form (0s and 1s) so that it can be stored, processed, and transmitted electronically.

Computers use internationally accepted standards like ASCII and Unicode for text and specific bit patterns for numbers and multimedia data. Data representation is essential because electronic circuits can understand only two states (ON/OFF), which are represented as 1 and 0.

Proper data representation makes complex real-world information understandable to digital systems and directly affects system performance, storage efficiency, and communication accuracy.

Types of Data Representation

1. **Numbers** : Numbers are converted into binary format so computers can perform arithmetic and logical operations.

Example : Decimal 5 → Binary 101

Different systems may use: Unsigned binary

Signed binary

Floating-point representation (for real numbers)

2. **Text** : Characters are represented using numeric character codes, which are then stored in binary.

Common standards : ASCII – 7 or 8 bits

Unicode – supports almost all world languages

Example: 'A' → ASCII 65 → Binary 01000001

3. **Images**: Images are stored as a grid of pixels. Each pixel is represented by binary values that define its color and brightness.

Black & white images: 1 bit per pixel

Color images: 8, 16, 24, or 32 bits per pixel

4. **Audio and Video** : Real-world sound and visuals are continuous signals. They are converted into digital form using sampling and quantization, producing sequences of binary numbers.

Examples:

Audio: MP3, WAV

Video: MP4, AVI, MKV

Video is essentially a series of digital images (frames) combined with synchronized digital audio.

❖ Types of Networks

The size and geographical coverage of a computer network determine its type. Networks can range from very small personal connections to large global systems. The main types of networks are PAN, LAN, CAN, MAN, and WAN.

1. PAN – Personal Area Network

A Personal Area Network (PAN) is the smallest network, used to connect devices belonging to a single person.



Key Features:

- Very short range (about 10 meters / 30 feet)
- Uses Bluetooth, infrared, NFC, or Zigbee
- Low power consumption

Advantages:

- Low cost and easy to set up
- Portable and flexible
- Low power consumption
- Good for personal devices (earbuds, phone, smartwatch)

Disadvantages:

- Very short range
- Slow data transfer compared to larger networks
- Limited number of devices
- Security can be weak if not configured properly

Examples:

- Mobile phone connected to wireless earphones
- Laptop connected to wireless mouse and keyboard

2. LAN – Local Area Network

A Local Area Network (LAN) connects devices within a limited area such as a home, office, school, or laboratory.



Key Features:

- High speed
- Low cost
- Privately owned
- Reliable communication

Advantages:

- High speed data transfer
- Low error rate
- Easy resource sharing (printers, files, internet)
- Relatively low cost

Disadvantages:

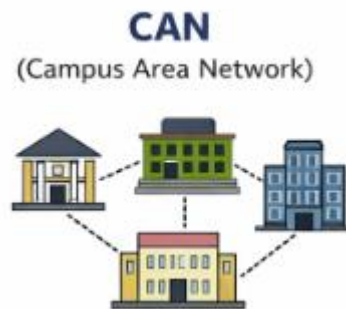
- Limited geographical area
- Needs maintenance and management
- Security risks if not protected
- Initial setup cost

Examples:

- Computer lab
- Office network
- Home Wi-Fi network

3. CAN – Campus Area Network

A Campus Area Network (CAN) connects multiple LANs within a campus or organization.



Key Features:

- Covers colleges, hospitals, or corporate campuses
- High-speed (often uses optical fiber)
- Owned and managed by a single organization

Advantages:

- High-speed connectivity across buildings
- Centralized management
- Cost-effective compared to MAN/WAN
- Better resource and data sharing

Disadvantages:

- More expensive than LAN
- Complex setup and maintenance
- Limited to a specific organization
- Requires skilled network administrators

Examples:

- University campus network
- Hospital network

4. MAN – Metropolitan Area Network

A Metropolitan Area Network (MAN) spans a city or large town and connects several LANs or CANs.



Key Features:

- Larger than LAN and CAN, smaller than WAN
- Medium to high speed
- Often managed by service providers

Advantages:

- Covers large geographical area
- Faster than WAN
- Useful for ISPs, cable TV, city-wide services
- Enables interconnection of many LANs and CANs

Disadvantages:

- High installation and maintenance cost
- More prone to security threats
- Complex management
- Performance depends on service providers

Examples:

- City-wide cable TV network
- Municipal internet network

5. WAN – Wide Area Network

A Wide Area Network (WAN) covers a very large geographical area, such as a country or the entire world.



Key Features:

- Uses leased lines, satellites, and fiber optics
- Expensive and complex
- Connects networks globally

Advantages:

- Very large coverage
- Enables global communication
- Supports cloud services and remote work
- High scalability

Disadvantages:

- Very expensive infrastructure
- Slower than LAN/MAN
- Higher latency and error rates

Examples:

- Internet
- Bank and airline reservation systems

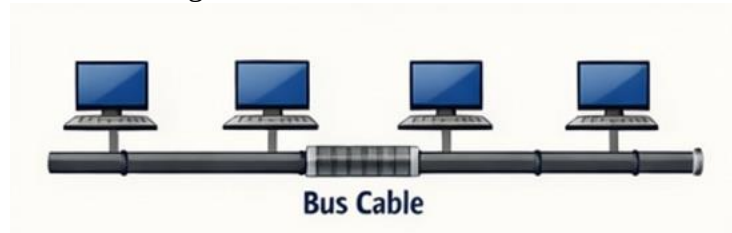
❖ Topologies

A network topology refers to the physical or logical arrangement of devices (nodes) and communication links in a computer network.

The main topologies are: Bus, Ring, Star, Mesh, Tree, and Hybrid.

1. Bus Topology

All devices are connected to a single central cable called the bus or backbone.



Features:

- Simple and low cost
- Easy to install
- If the main cable fails, the entire network fails
- Performance decreases as more devices are added

Advantages:

- Easy to set up and understand.
- Requires less cabling than other topologies.
- Works well in small networks with limited devices.

Disadvantages:

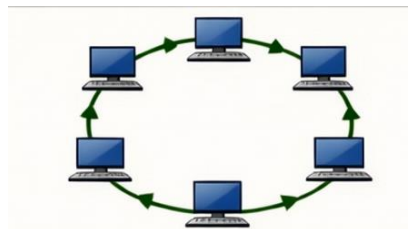
- If the main cable (bus) fails, the entire network is affected.
- Performance degrades as more devices are added.
- Finding faults or locating failures can be challenging.

Example:

- Early Ethernet networks.

2. Ring Topology

Each device is connected to two other devices, forming a circular loop. Data travels in one direction (or both in dual ring).



Features:

- Equal access for all nodes
- Data passes through each device
- Failure of one node/cable can break the network
- Slower than star topology

Advantages:

- Data travels in one direction, reducing collisions.
- Easy to install and manage.
- No single point of failure like a central hub.

Disadvantages:

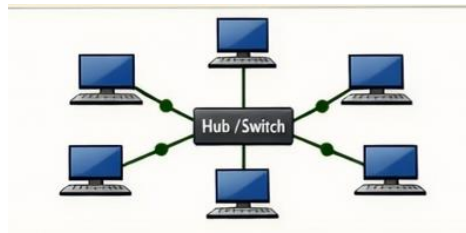
- If one device or connection fails, the entire network can be disrupted.

Example:

- Token Ring networks.

3. Star Topology

All devices are connected to a central hub or switch.



Features:

- Easy to install and manage
- Failure of one cable affects only one device
- Failure of central device stops the whole network
- Most commonly used topology today

Advantages:

- Easy to manage and troubleshoot.
- Easy to add or remove devices without affecting the rest of the network.
- Failure of one device does not affect others.

Disadvantages:

- If the hub fails, the whole network can go down.
- Requires more cabling compared to bus topology.
- Managing large star topologies can become complex.

Example:

- Modern LANs, office networks.

4. Mesh Topology

Each device is connected to many or all other devices.



Features:

- Very high reliability
- Multiple data paths
- Expensive and complex
- Used where failure is not acceptable

Advantages:

- High reliability; reroutes data if a link fails.
- Excellent fault tolerance.

Disadvantages:

- Expensive due to extensive cabling.
- Complex setup and management.

Example:

- Military networks, internet backbone.

5. Tree Topology

A hierarchical structure combining characteristics of bus and star topologies.



Features:

- Easy to expand
- Central root with branches
- Failure of backbone affects many nodes

Advantages:

- Scalable and easy to manage.
- Good for larger networks.

Disadvantages:

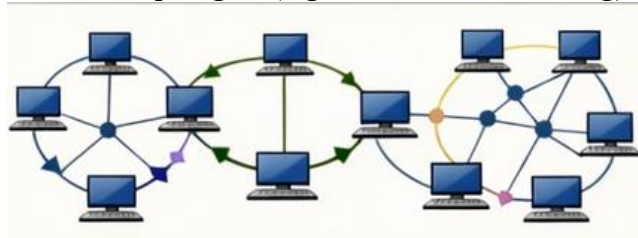
- Root node failure affects all connected segments.
- More complex than star or bus topology.

Example:

- Large organizations, school networks.

6. Hybrid Topology

A combination of two or more topologies (e.g., star + bus, star + ring).



Features:

- Flexible and scalable
- Reliable
- Complex design
- Costly to implement

Advantages:

- Flexible and adaptable.
- Optimizes strengths of various topologies.

Disadvantages:

- Increased complexity in management.
- Potentially high costs.

Example:

- Most real-world enterprise networks.

❖ Key terms

1. Message

A message is the information or data that is to be communicated from one device to another. It can be in the form of: text, numbers, images, audio, or video.

Example: An email, a photo, a voice note.

2. Sender

The sender is the device that creates and transmits the message.

Examples: Computer, mobile phone, server, sensor.

The sender converts the data into signals before transmission.

3. Receiver

The receiver is the device that receives and interprets the message.

Examples: Computer, smartphone, printer.

The receiver converts the signals back into meaningful data.

4. Medium (Transmission Medium)

The medium is the physical or wireless path through which data travels from sender to receiver.

Types of medium:

- Guided (wired): Twisted pair cable, coaxial cable, optical fiber
- Unguided (wireless): Radio waves, microwaves, infrared

Example: Internet cables, Wi-Fi signals.

5. Jitter

Jitter is the variation in the arrival time of data packets at the receiver.

High jitter causes distortion in audio/video calls and live streaming.

Example: Breaking voice or frozen video in a video call.

6. Types of Data

Data in communication systems can be represented in different forms:

- Text – letters, symbols
- Numbers – numeric values
- Images – pictures, graphics
- Audio – sound, music, speech
- Video – moving visuals

All types of data are finally converted into binary (0s and 1s).

❖ Network Model

A network model is a layered framework that explains how data is transmitted from one computer to another over a network. Each layer has a specific function, which makes network design, implementation, and troubleshooting easier.

The two most important network models are:

1. OSI Model (Open Systems Interconnection)
2. TCP/IP Model (Transmission Control Protocol / Internet Protocol)

1. OSI Model (Open Systems Interconnection)

The OSI (Open Systems Interconnection) model is a conceptual reference model developed by the International Organization for Standardization (ISO) to standardize communication between different computer systems. It divides the process of network communication into seven layers, where each layer performs a specific function. This layered approach makes network design, implementation, and troubleshooting easier.



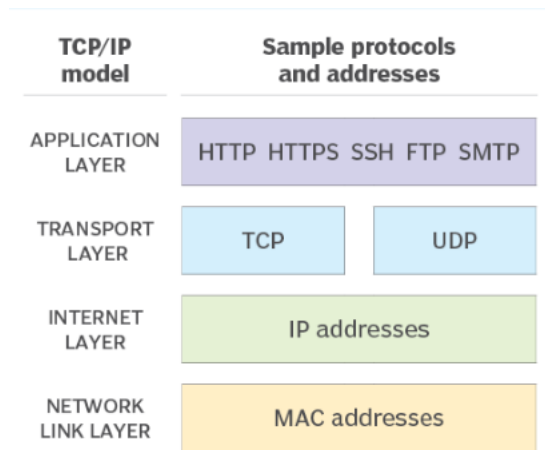
Seven Layers of the OSI Model

1. Physical Layer
It is responsible for the transmission of raw bits (0s and 1s) over the physical medium. It defines cables, connectors, voltage levels, and data rates.
2. Data Link Layer
It provides node-to-node delivery, framing, physical (MAC) addressing, and error detection and correction.
3. Network Layer
It handles logical addressing and routing, selecting the best path for data transmission.
Example: IP addressing.
5. Transport Layer
It ensures end-to-end delivery, error control, flow control, and data segmentation.
Example protocols: TCP, UDP.
6. Session Layer
It establishes, manages, and terminates sessions between communicating devices.
7. Presentation Layer
It performs data translation, encryption/decryption, and compression, ensuring data is in a readable format.
8. Application Layer
It provides network services to users, such as email, file transfer, and web services.
Examples: HTTP, FTP, SMTP.

2. TCP/IP Model (Transmission Control Protocol / Internet Protocol)

The TCP/IP (Transmission Control Protocol / Internet Protocol) model is a practical network model developed by the U.S. Department of Defense (DARPA). It forms the foundation of the Internet and is used for real-world data communication. The TCP/IP model explains how data is transmitted from one device to another using a layered architecture.

TCP/IP model consists of four layers, each responsible for specific network functions.



1. Application Layer
 - Description of the Application Layer.
 - It provides network services directly to applications, such as web browsers or email clients.
 - High-level protocols and applications.
 - Example Protocols: HTTP, FTP, DNS, SMTP.
2. Transport Layer
 - Description of the Transport Layer.
 - This layer is responsible for end-to-end communication between applications.
 - Example Protocols: TCP, UDP.
3. Internet Layer
 - Description of the Internet Layer.
 - This layer deals with logical addressing and routing of data packets between different networks.
 - Routing and addressing (IP addresses).
 - Example Protocols: IP, ICMP, IGMP.
4. Link Layer (Network Access Layer)
 - Description of the Link Layer in TCP/IP.
 - This is the lowest layer in the TCP/IP model and is concerned with the physical transmission of data between devices on the same network.
 - Physical network components (Ethernet, Wi-Fi).
 - Example Protocols: Ethernet etc.

○ **Comparison of OSI and TCP/IP Models.**

Feature	OSI Model	TCP/IP Model
Full form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Developed by	ISO (International Organization for Standardization)	DARPA (U.S. Department of Defense)
Number of layers	7 layers	4 layers
Nature	Conceptual / reference model	Practical / implementation model
Layer separation	Very clear and strict	Less strict, more flexible
Session & Presentation layers	Separate layers	Included in Application layer
Network support	Protocol-independent	Protocol-oriented
Usage	Teaching, design, troubleshooting	Internet and real networks
Development approach	Model first, protocols later	Protocols first, model later
Popularity	Educational and theoretical	Widely implemented worldwide

❖ **IMP QUESTIONS :**

- 1. OSI Model explanation**
- 2. OSI vs TCP/IP comparison**
- 3. Network Topologies with diagrams**
- 4. Types of Networks**
- 5. Components of Data Communication**
- 6. TCP/IP Model layers**

